

SUMIT YADAV

Mechanistic Interpretability | AI Alignment & Safety | NLP

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RESEARCH STATEMENT

I work at the intersection of **mechanistic interpretability**, **AI alignment**, and **robustness**. My research studies how safety-aligned LLMs fail silently, through over-refusals, geometric misrepresentation, and surface-level triggers, and how the internal structure of representations can be understood and steered to make models safer and more reliable. Much of my strongest work is sole-authored: I combine representation-space analysis, adversarial probing, and systems engineering into empirically grounded methods. My work has been accepted at **ACL 2026 (CORE A*)**, and I am especially interested in extending interpretability and safety to multilingual and low-resource settings.

SELECTED RESEARCH & PUBLICATIONS

- SafeConstellations: Mitigating Over-Refusals in LLMs Through Task-Aware Representation Steering** ACL 2026 Main (CORE A*) 2026
- LLMs trace consistent, task-specific “*constellation*” trajectories in embedding space that shift predictably between refusal and non-refusal states.
 - SafeConstellations** is an inference-time, task-aware trajectory-shifting method that nudges representations toward non-refusal pathways *only* on over-refusal-prone tasks, cutting over-refusals by up to **73%** with minimal utility loss and no retraining. (*with U. Maskey, M. Dras, U. Naseem.*)
- Geometric Phases of Mechanism Formation in Neural Networks** (*sole-authored*) 2026
- GLOW @ IJCAI-ECAI 2026 (Best Paper Candidate)
- Using linear probes and CKA across dense training checkpoints, shows classification mechanisms form **output-layer-first** and within the first ~5% of training (Cohen’s $d = 3.68$); the deep-first pattern holds in from-scratch LLM pretraining (GPT-2 Small, SmoLM2-135M) and reproduces on public Pythia and OLMo-2 checkpoints.
- MaiBERT: A Pre-training Corpus and Language Model for Low-Resourced Maithili** 2026
- LoResLM @ EACL 2026
- First monolingual BERT for Maithili (~50M speakers); **87.02%** news-classification accuracy, outperforming multilingual baselines (MuRIL, NepBERTa), addressing fairness and access for underrepresented communities.
- How Far Can a Single Vector Carry a Language? Mechanistic Limits of Inference-Time Steering for Low-Resource Devanagari Languages** (*first author*) 2026
- Under review, Machine Learning (Springer)
- With **no fine-tuning**, redirects a model’s **Hindi** representations toward unsupported Devanagari languages (Nepali, Maithili, Bhojpuri). Finds a shared **hourglass geometry** and an **adherence–fluency frontier** that bounds single-direction steering: adherence arrives before fluency, and **separability rather than proximity** predicts steerability (Hindi 3.23 vs English 1.37). Unifies the SafeConstellations and maiBERT lines.
- On the Relationship Between Representation Geometry and Generalization** (*sole-authored*) 2026
- Pre-Print 2026
- Establishes that a label-free geometric metric, **effective dimension**, strongly predicts generalization across vision and language (partial $r = 0.75$ over 52 ImageNet models, 13 architectures; carries to NLP encoders and decoder-only LLMs). The link is causal: degrading the geometry hurts accuracy while PCA cleanup preserves it, a diagnostic relevant to distribution-shift detection and robustness.
- The Steerability Spectrum of Visual Attributes** ongoing 2026
- Measures how readily different visual features (size, distance, orientation) can be steered inside vision models (DINOv2, CLIP, SigLIP): some lie on a single clean, controllable direction while others are distributed and resist steering, probing what accounts for the difference.
- Also: **Revolutionizing Currency Security with YOLOv8** (J. Bus. Econ. Stud., 2024; 0.986 true-positive recall) · **Machine Learning Analysis of Tirhuta Lipi** (2023) · **Support Vectors for Imbalanced Text Classification** (2023).

EXPERIENCE

- Astha.ai** USA (Remote)
- AI Researcher, AI Security & Agentic Systems* May 2025 – Present
- Research-to-production work in AI security: **primary author** of the SAFE-MCP vulnerability scanner (**78+** MITRE ATT&CK-mapped techniques covering tool poisoning and injected tool descriptions) and **lead engineer** on a Go-based Kubernetes compliance agent running both inside customer clusters and in the SaaS platform.
 - Recurring research thread — **making LLM-based verdicts reproducible**: a statistical analysis mode for the MCP scanner and a repeated-run consensus mechanism with fixed sampling parameters for compliance verdicts, alongside LLM-driven scan planning over a catalog of ~**470** compliance-control skills and prompt-injection hardening with tree-sitter semantic chunking.
- AMNIL Technologies** Nepal (Hybrid)
- AI Engineer, RAG & Infrastructure* Jun 2024 – May 2025
- Guardrails + **LLM-as-a-Judge** evaluation; hybrid-search RAG (Qdrant, +35% retrieval accuracy); self-hosted LLM serving with vLLM (40% lower latency).

GradeUp Educations

Data Team Lead

Kathmandu, Nepal

Jan 2022 – Jun 2024

- Led the data team: learning agents/chatbots, an automated grade-evaluation system, and semantic-similarity matching (NLP + image processing).

DeepLearning.AI

GAN Specialization Mentor

Global (Remote)

Aug 2021 – Present

- Technical mentor for the GANs Specialization (loss functions, training stability, generative-model robustness).

SELECTED PROJECTS & OPEN SOURCE

Removing the “Workspace” Layers of Llama: interpretability study locating a decodable-yet-entangled middle band of layers in Llama-3.1-8B; ablating the whole band collapses reasoning while shallow recall survives (linear probes + reversible layer-ablation, nnsight/NDIF to 405B). [code].

World Models (toy experiments): built Dreamer-style world models (RSSM: GRU + stochastic latents) from scratch for continuous and visual control; agents trained purely in imagination transfer to the real task (CartPole 500/500), and short-horizon dreams with periodic reality-grounding cut drift $11\times$. [code].

SAFE-MCP: community security framework for the Model Context Protocol (ATT&CK-style threat/defense rules; prompt-injection & tool-poisoning modules).

Telebot-Claude-Bridge: permission-gated, human-in-the-loop remote control of the Claude Code CLI (inline allow/deny per tool call, audit logging).

Maithili speech: open Whisper-tiny **ASR** and MMS-TTS **TTS** for Maithili (live HF Spaces demos); **maiBERT**, the first open BERT for Maithili (HF).

NPL Coders: founder of Nepal’s national data-science competition community.

TECHNICAL PROFICIENCY

Interpretability & Alignment: representation & activation steering, mechanistic interpretability, linear probes / CKA, representation geometry, effective dimension, over-refusal mitigation.

LLMs & NLP: LLMs (Llama, Claude, GPT), fine-tuning (PEFT/LoRA), RAG, multilingual / low-resource NLP, tokenizer training, embedding alignment.

Agentic & Security: Model Context Protocol, multi-agent frameworks, MITRE ATT&CK mapping, prompt-injection & tool-poisoning defense, adversarial evaluation.

Tools: Python (expert), PyTorch, TensorFlow, vLLM, FastAPI, Docker, Kubernetes, vector DBs (Qdrant, ChromaDB), nnsight.

Research methods: empirical NLP evaluation, information-theoretic metrics, adversarial benchmarking, statistical analysis.

EDUCATION

Pulchowk Engineering College, IOE, Tribhuvan University

Kathmandu, Nepal

Bachelor of Computer Engineering

2019 – 2024

- B.E. thesis: “Evaluating Auto-Encoding Transformer Language Models for Maithili Text Classification”: my first Maithili masked language model, and the foundation for maiBERT.
- Key subjects: Artificial Intelligence, Data Structures & Algorithms, Distributed Systems, Big Data Technologies, Network Security, Compiler Design.

HONORS & AWARDS

Best Paper Candidate, GLOW @ IJCAI-ECAI 2026 (sole-authored) · Outstanding Reviewer, Mechanistic Interpretability Workshop (ICML 2026) · Winner, GritFeat AI Hackathon (2023) · 1st Runner-Up, Locus Dataverse (2023) · Best AI Project, DELTA 3.0 (2023) · Winner, IT-Meet Image Challenge (2022) · Winner, LogPoint Capture The Flag (2022)